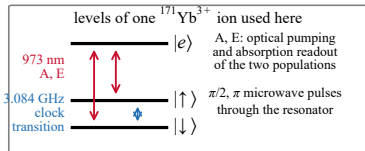
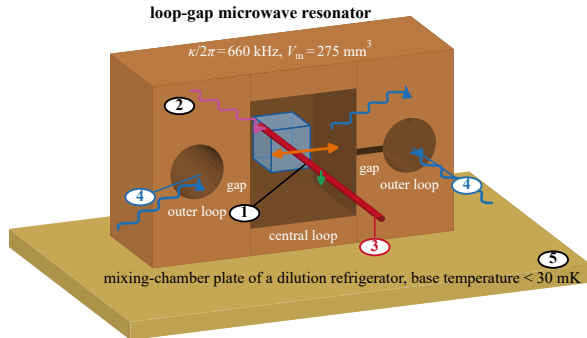


(a)

**Key to (a), loop-gap device as built [Fukumori *et al.*]:**

- ① $^{171}\text{Yb}^{3+}$: CaWO₄ crystal, $4.4 \times 4.6 \times 5 \text{ mm}$, 4.96 ppm ^{171}Yb , glued in the central loop, c axis (polished faces) along the beam
- ② loop-gap resonator: central loop, two outer loops, narrow gaps;
 $\kappa/2\pi = 660 \text{ kHz}$, $V_m = 275 \text{ mm}^3$, $g/2\pi = 15 \text{ mHz}$
- ③ 973 nm optical I/O (fibre collimators): pumping into $|\downarrow\rangle$, absorption readout of $|\uparrow\rangle, |\downarrow\rangle$; the beam covers about 1% of the crystal face
- ④ microwave I/O antennas at the outer loops: $\pi/2, \pi$ pulses; transmission S_{21} calibrates N_0
- ⑤ mixing-chamber plate, base $< 30 \text{ mK}$; spin temperature 80 mK

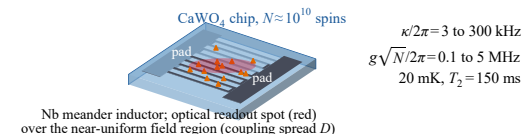
Spin and operating parameters:

$\omega_c/2\pi = 3.084 \text{ GHz}$ (zero-field clock transition), $\gamma_{\text{inh}}/2\pi < 5 \text{ kHz}$, $T_2 > 150 \text{ ms}$
 $\Delta/2\pi = 22 \text{ MHz}$, $2\Delta/\kappa = 67$, $N_0 \leq 7 \times 10^{14}$, $n_{\text{th}} = 0.19$ at 80 mK
 model uses $\gamma_{\text{inh}}/2\pi = 5 \text{ kHz}$ (Voigt, 30% Lorentzian) and $T_2 = 150 \text{ ms}$

Processes kept in the model (glyphs at the crystal):

- orange diamond: one-axis twisting $\chi \hat{J}_+ \hat{J}_-$ (cavity-mediated exchange)
- blue wavy arrow: collective emission through the detuned cavity, Γ_{SR}
- pink wavy arrow: thermal cavity photons absorbed collectively, $\Gamma_{\text{SR}} n_{\text{th}}$
- green arrow: single-spin dephasing, $1/T_2$

(b) planar superconducting resonator on the same crystal: the design-map case (proposed, not built)



(c)

